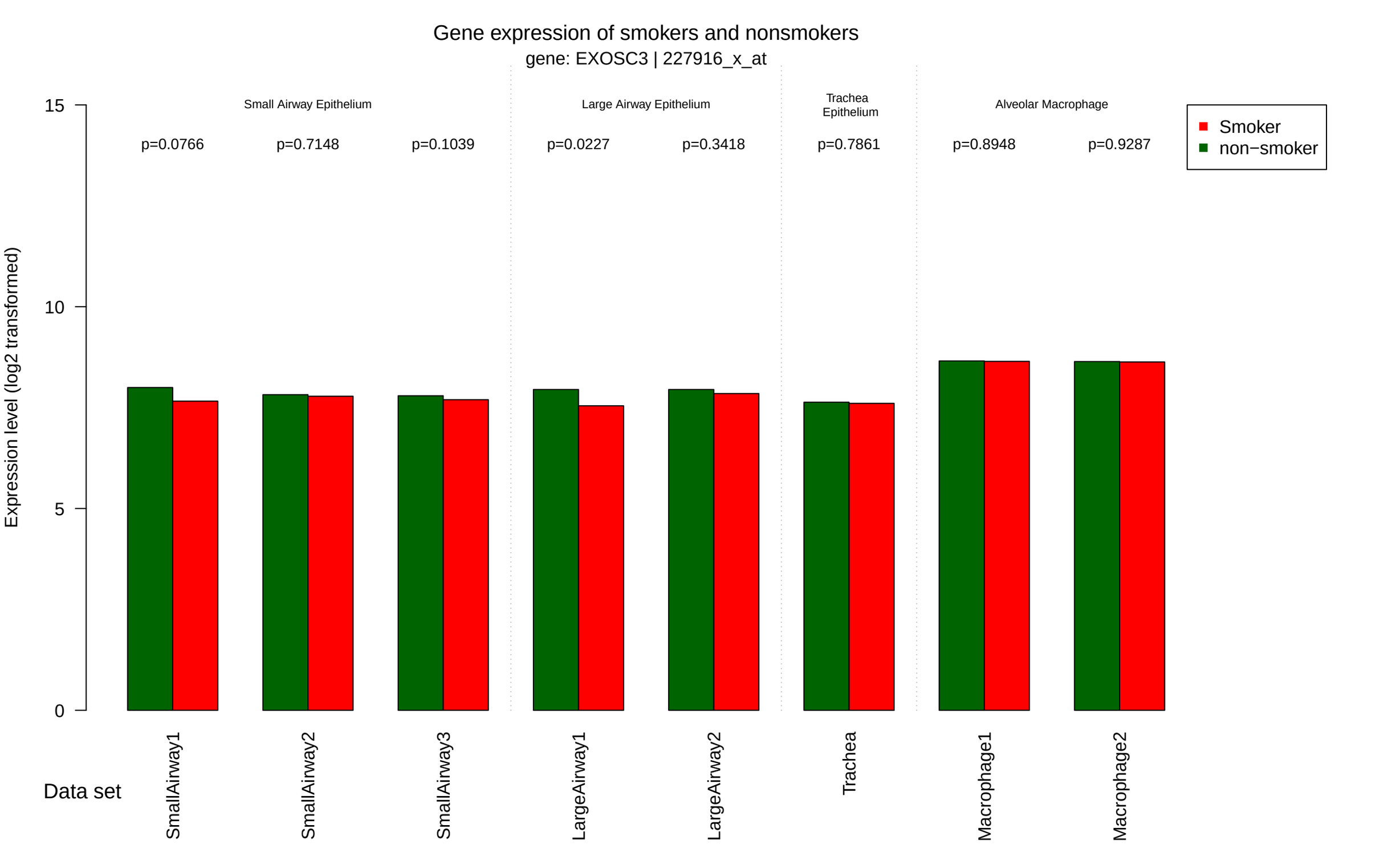
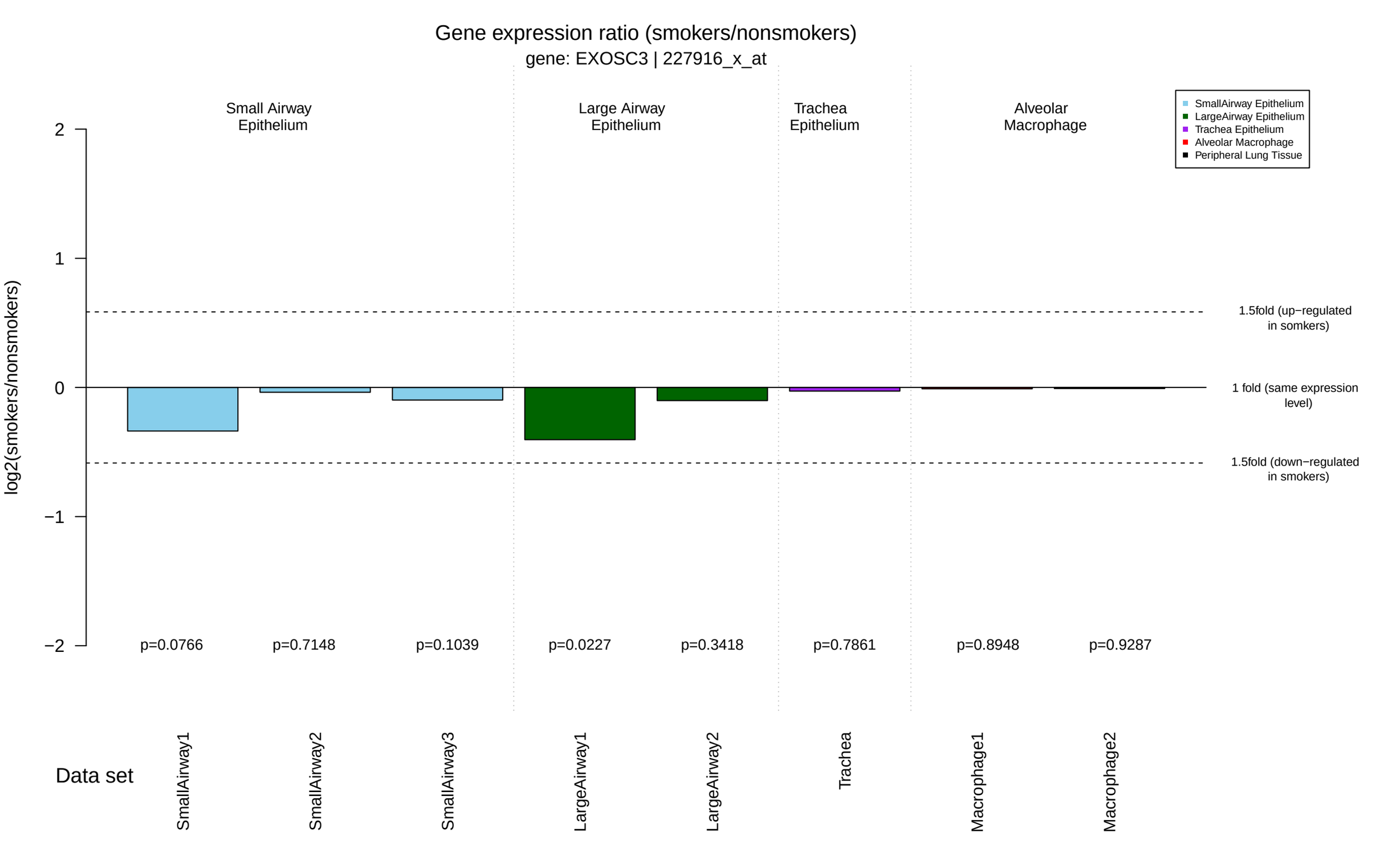
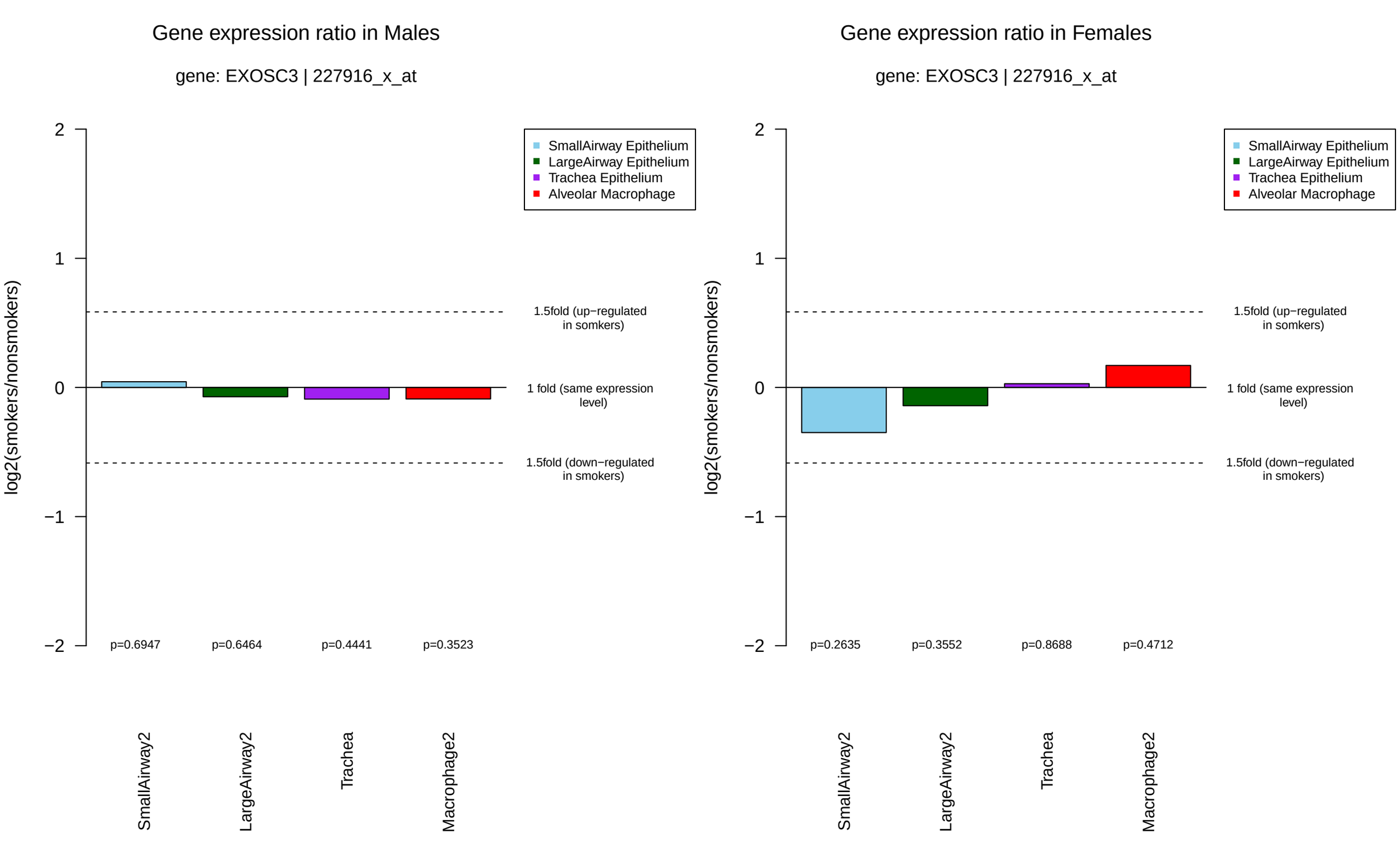
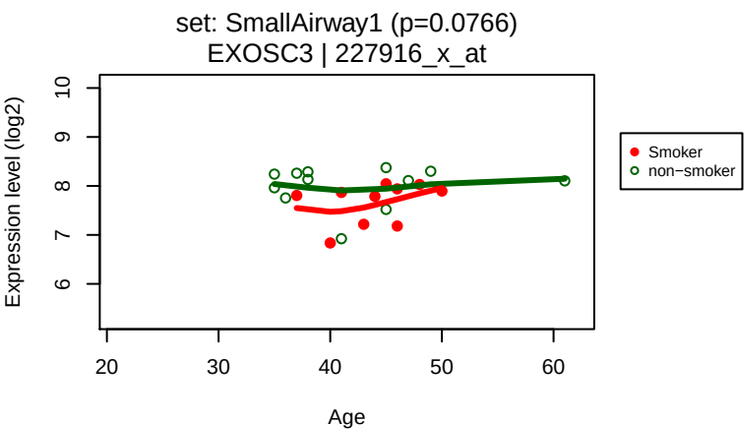
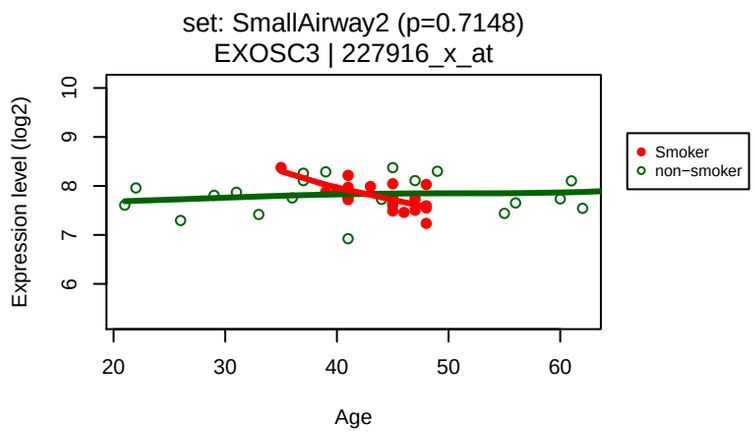
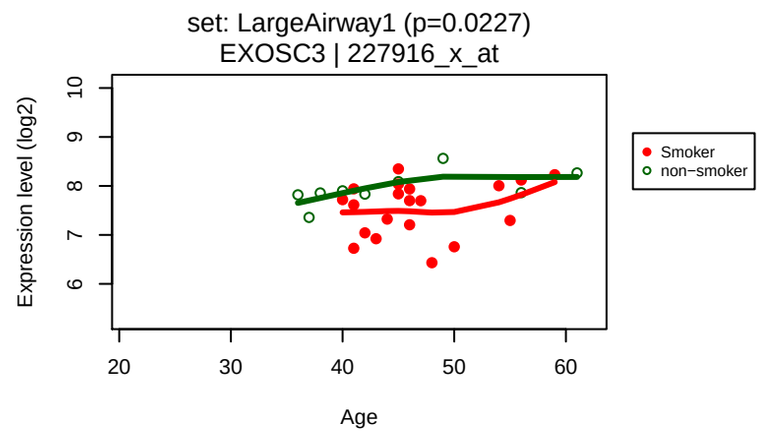

Gene expression of smokers and nonsmokers
gene: EXOSC3 | 227916_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0766
p=0.7148
p=0.1039
p=0.0227
p=0.3418
p=0.7861
p=0.8948
p=0.9287
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

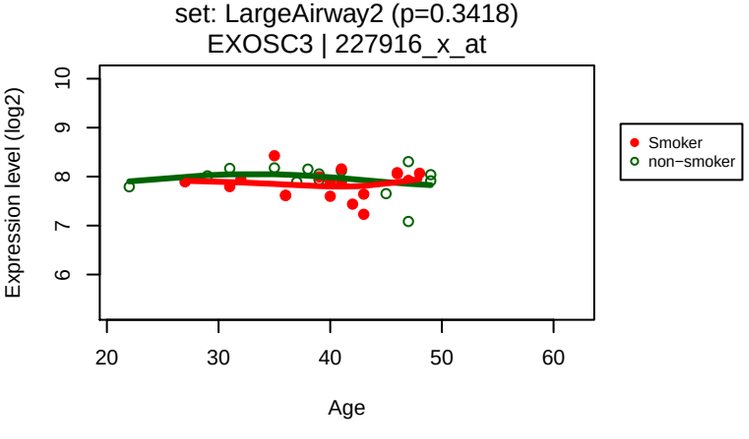

Gene expression ratio (smokers/nonsmokers)
gene: EXOSC3 | 227916_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0766
p=0.7148
p=0.1039
p=0.0227
p=0.3418
p=0.7861
p=0.8948
p=0.9287
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

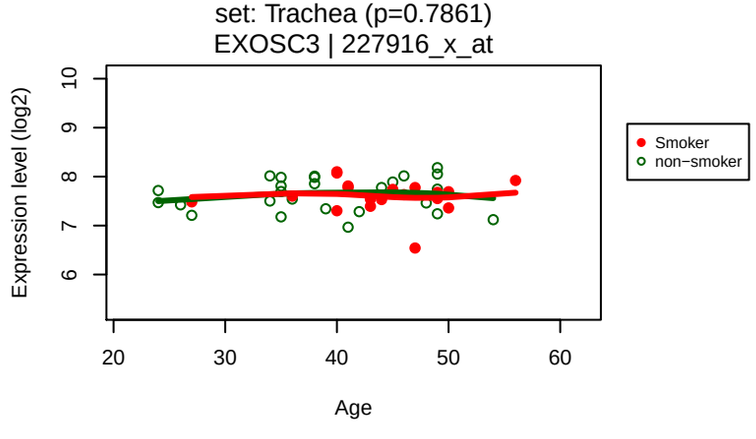

Gene expression ratio in Males
gene: EXOSC3 | 227916_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6947
p=0.6464
p=0.4441
p=0.3523
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: EXOSC3 | 227916_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2635
p=0.3552
p=0.8688
p=0.4712
SmallAirway2
LargeAirway2
Trachea
Macrophage2

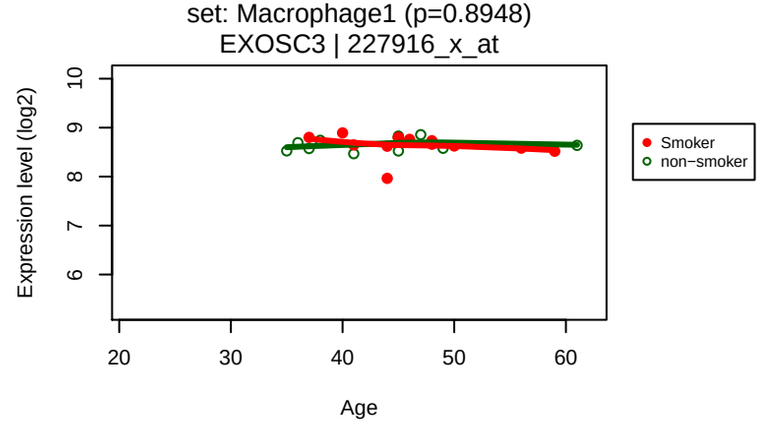

set: SmallAirway1 (p=0.0766)
EXOSC3 | 227916_x_at
Expression level (log2)
Age
Smoker
non−smoker

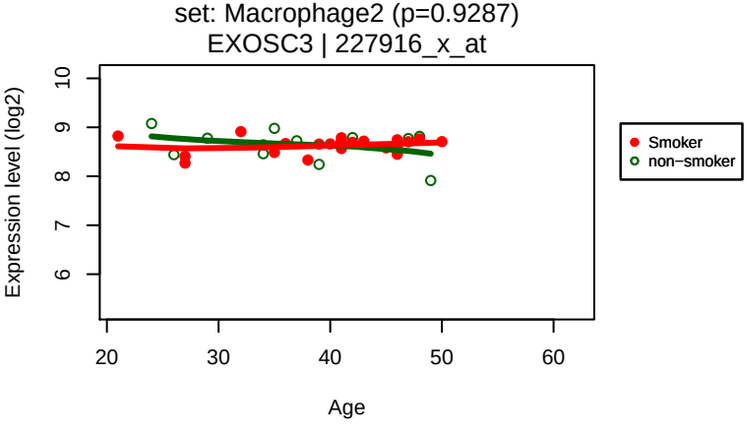

set: SmallAirway2 (p=0.7148)
EXOSC3 | 227916_x_at
Expression level (log2)
Age
Smoker
non−smoker

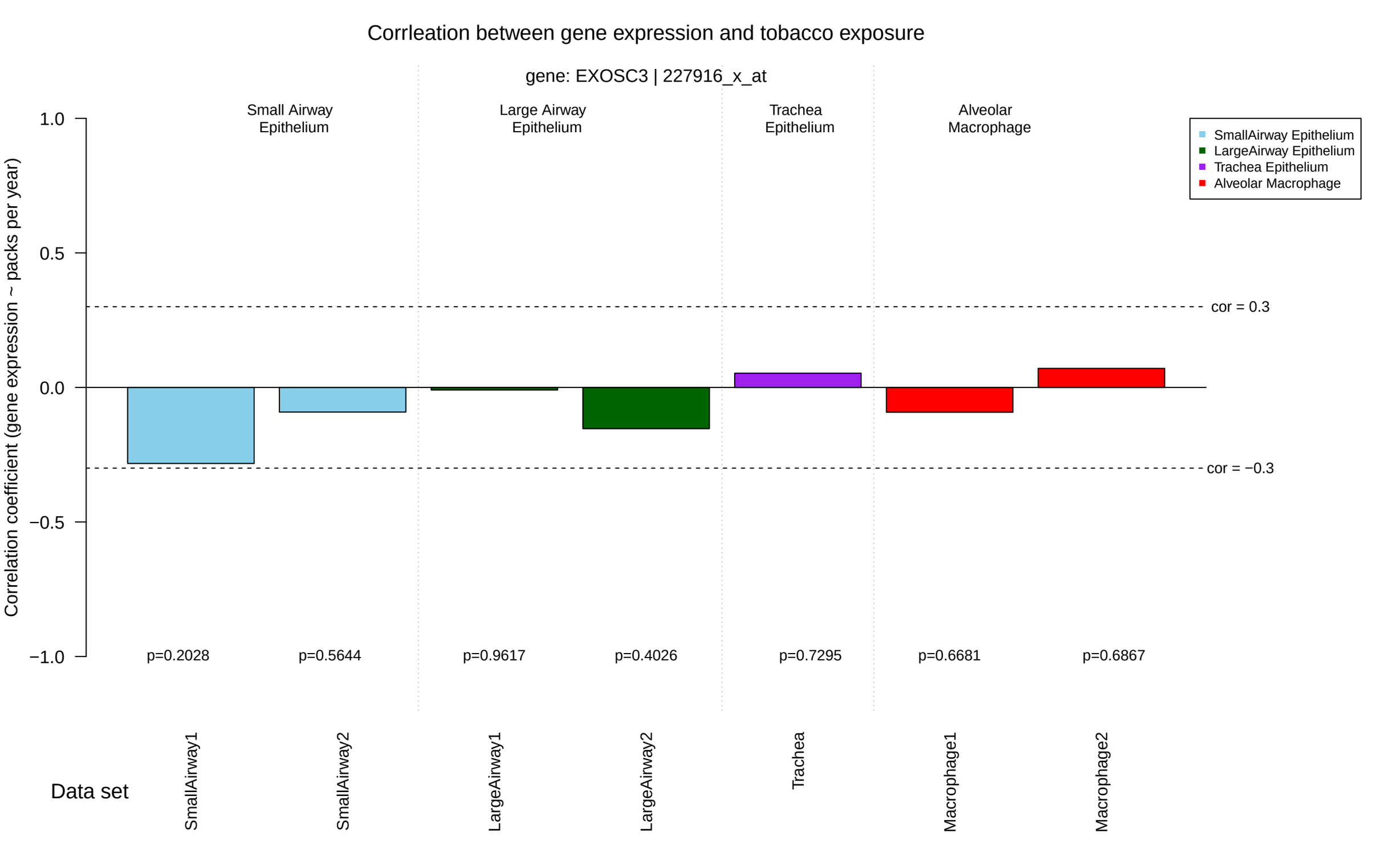

set: LargeAirway1 (p=0.0227)
EXOSC3 | 227916_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3418)
EXOSC3 | 227916_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7861)
EXOSC3 | 227916_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8948)
EXOSC3 | 227916_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9287)
EXOSC3 | 227916_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: EXOSC3 | 227916_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2028
p=0.5644
p=0.9617
p=0.4026
p=0.7295
p=0.6681
p=0.6867
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2